Introduction

Priority Queue : using an unordered list
Data Structures and its Applications (UE25CS52A)

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Quiz : Priority Queue : using an unordered list

1. Which statement best describes enque in a priority queue implemented using an unordered list?

a) The number of operations always proportional to the number of elements in the priority queue
b) The number of operations in the worst case proportional to the number of elements in the priority queue
c) Always a few operations not depending on the number of elements in the priority queue
d) Sometimes a few operations not depending on the number of elements in the priority queue

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Quiz : Priority Queue : using an unordered list

2. Which statement best describes deque in a priority queue implemented using an unordered list?

a) The number of operations always proportional to the number of elements in the priority queue
b) The number of operations in the worst case proportional to the number of elements in the priority queue
c) Always a few operations not depending on the number of elements in the priority queue
d) Sometimes a few operations not depending on the number of elements in the priority queue

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Quiz : Priority Queue : using an unordered list

2. Which statement best describes deque in a priority queue implemented using an unordered list?

a) The number of operations always proportional to the number of elements in the priority queue
b) The number of operations in the worst case proportional to the number of elements in the priority queue, a few operations otherwise
c) Always a few operations not depending on the number of elements in the priority queue
d) Sometimes a few operations not depending on the number of elements in the priority queue

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Quiz : Priority Queue : using an unordered list

3. In an unordered-list implementation of a priority queue, how are elements stored?

a) Always in increasing priority order
b) Always in decreasing priority order
c) Without maintaining any priority order
d) In FIFO order

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Quiz : Priority Queue : using an unordered list

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Quiz : Priority Queue : using an unordered list

4. If insertion is always performed at the head of a singly linked list having n nodes, what is the # of operations on insertion in the priority queue?

a) A few operations not depending on n
b) # of operations proportional to n
c) # of operations proportional to n ^ 2
d) # of operations proportional to log(n) to base 2

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Quiz : Priority Queue : using an unordered list

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Quiz : Priority Queue : using an unordered list

5. Which operation is not efficient in a priority queue implemented using a list?

a) enque
b) deque
c) initialize
d) Check for empty

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Priority Queue : using an unordered list

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5. Which operation is not efficient in a priority queue implemented using a list?

a) enque
b) deque
c) initialize
d) Check for empty

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Priority Queue: concept

- A priority queue is an abstract data type in which each element is associated with a priority.
- A priority queue removes the component with the highest priority first.
- A priority is a number associated with each element
- Higher the numeric value, could mean higher the priority.
- Or lower the numeric value, could mean higher priority

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Priority Queue: some examples

- Patient treatment
- Scheduling of processes

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Priority Queue: interface

initialize() deinitialize()	Create an empty queue Make the queue empty
enqueue(x) I	Insert x We may add in number of ways. This in turn would affect dequeue
dequeue()	Remove the element with the highest priority. This in turn depends how enqueue operation was done
isEmpty()	Check whether the queue is empty
isFull()	Check whether the queue is full

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Priority Queue: Possible Implementations

- Unodered list
- Ordered list : ordered based on priority
- Unordered array
- Ordered array : ordered based on priority
- Heap tree

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Priority Queue: unordered list

- Implemented using queue implementation
- Could also implement using a list

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Priority Queue: unordered list implementation

- Initialize : make the front and the rear NULL
- Deinitialize : deallocate the nodes and then make the front and the rear NULL
- Is_empty : check front is NULL
- Is_full : check whether memory allocation fails

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Priority Queue: unordered list implementation

- Enque :
  - Make a node
  - Populate
  - Add at the rear
  - Change front if required
  - A few operations
- Deque :
  - Traverse from front to rear
  - Find the position of the leftmost highest priority node
  - Copy the job to return
  - Suitably delete that node

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Priority Queue: Applications of priority queue :

- CPU Scheduling of processes
- Hospital Emergency Services
- Printer Scheduling
- Network Packet Scheduling
- Dijkstra's Shortest Path Algorithm
- Prim's Minimum Spanning Tree Algorithm
- Event-Driven Simulation
- Huffman Coding

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THANK YOU

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